Task-3 Regression Model for Prediction Using Linear Regression

1. Introduction

The objective of this project is to build a regression model for predicting the selling price of T-shirts using the Myntra T-Shirt Dataset. Linear Regression was used to establish a relationship between the original price, discount, and selling price.

2. Dataset Description

The dataset consists of 4,950 T-shirt products.

Attributes Used

- Price
- Discount
- Selling Price

Target Variable:

- Selling Price

3. Data Preprocessing

The following preprocessing steps were performed:

- Missing value handling
- Duplicate removal
- Conversion of Price and Discount into numeric format

df.dropna(inplace=True)

4. Regression Model

A Linear Regression model was selected because it is simple and suitable for predicting continuous numerical values.

from sklearn.linear_model import LinearRegression

5. Training and Testing

Dataset split:

- Training Set: 80%
- Testing Set: 20%

```
train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

6. Model Evaluation

R^2 Score

Measures how well the model explains the variance in the target variable.

```
r2_score(y_test, y_pred)
```

```
R2 Score : 0.7993926221256538
```

Mean Squared Error (MSE)

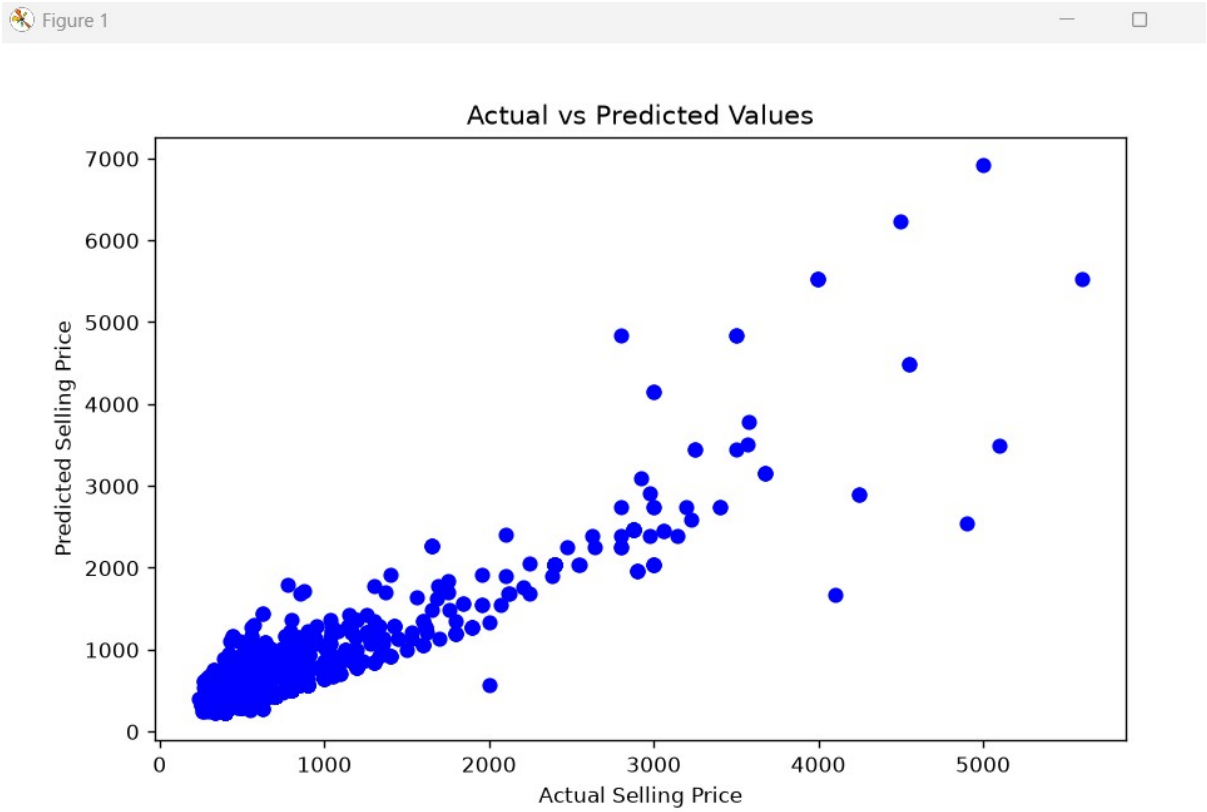
Measures prediction error.

```
mean_squared_error(y_test, y_pred)
```

```
Mean Squared Error : 119752.04957761306
```

7. Visualization

Actual vs Predicted Values



Observation: The scatter plot shows the relationship between predicted and actual selling prices. Points closer to the diagonal indicate better prediction accuracy.

8. Results

Metric	Value
R^2 Score	R2 Score : 0.7993926221256538
Mean Squared Error	Mean Squared Error : 119752.04957761306

9. Conclusion

A Linear Regression model was successfully developed to predict T-shirt selling prices. The model was trained using original price and discount information. Performance was evaluated using R^2 Score and Mean Squared Error. The visualization demonstrated the relationship between actual and predicted values, showing the effectiveness of the regression model.